$$H_k^{2n} = \frac{1}{2^{n+1}\sqrt{\pi}} (\sin\phi)^n P_{k-n}^{(n,n)}(\cos\phi)\cos(n\theta)$$

$$\|H_k^{2n}\|_{L^2(S^2)} = \int_0^{2\pi}\int_0^{\pi} |H_k^{2n}|^2 \sin\phi \, d\phi d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \frac{1}{2^{2n+2}\pi} (\sin\phi)^{2n} \left(P_{k-n}^{(n,n)}(\cos\phi)\right)^2 \cos^2 n\theta \sin\phi \, d\phi \, d\theta$$

$$= \frac{1}{2^{2n+2}\pi} \int_0^{2\pi} \cos^2(n\theta) d\theta \times \int_0^{\pi} (\sin\phi)^{2n+1} \left(P_{k-n}^{(n,n)}(\cos\phi)\right)^2 d\phi$$

$$= \int_0^{2\pi} \cos^2(n\theta) d\theta = \int_0^{2\pi} \frac{1-\cos(2n\theta)}{2} d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} d\theta - \frac{1}{2}\int_0^{2\pi} \cos(2n\theta) d\theta$$

$$= \pi - \frac{1}{2} \times \frac{1}{2n} \times \sin(2n\theta)\Big|_0^{2\pi}$$

zero

$$= \pi$$

$x = \cos\phi$

$dx = -\sin\phi d\phi$

$x = \cos 0 = 1$

$x = \cos\pi = -1$

$$= \frac{\pi}{2^{2n+2}\pi} \times \int_0^{\pi} (\sin\phi)^{2n} \left(P_{k-n}^{(n,n)}(\cos\phi)\right)^2 \sin\phi \, d\phi$$

$$= \frac{-1}{2^{2n+2}} \int_{+1}^{-1} (1-x^2)^n \left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

$$= \frac{1}{2^{2n+2}} \int_{-1}^{+1} (1-x)^n (1+x)^n \left(P_{k-n}^{(n,n)}(x)\right)^2 dx$$

The Jacobi polynomials satisfy the orthogonality condition

$$\int_{-1}^{1} (1-x)^{\alpha}(1+x)^{\beta} P_n^{(\alpha,\beta)}(x) P_m^{(\alpha,\beta)}(x)\, dx = \frac{2^{\alpha+\beta+1}\Gamma(\alpha+n+1)\Gamma(\beta+n+1)}{n!\,(\alpha+\beta+1+2n)\,\Gamma(\alpha+\beta+n+1)}\delta_{mn}, \tag{2.7}$$

$\alpha = \beta = n$

$n = m = k-n$

$$= \frac{1}{2^{2n+2}} \times \frac{2^{2n+1}\,\Gamma(k+n+1)^2}{(k-n)!\;(n+n+1+2k-2n)\,\Gamma(n+n+k-n+1)}$$

$$= \frac{1}{2} \times \frac{\left(\Gamma(k+1)\right)^2}{(k-n)!\;(2k+1)\,\Gamma(n+k+1)}$$

$$= \frac{(k!)^2}{2(2k+1)(k-n)!\;(n+k)!}$$

$$H_k^{2n+1} = \frac{1}{2^{n+1}\sqrt{\pi}}$$

$$\left\| H_k^{2n+1} \right\|_2^2 = \frac{1}{2^{n+2}\pi} \int_0^{2\pi} \sin^2(n\theta)\, d\theta = \pi$$

$\sin(n\theta)$

$$\times \int_0^{\pi} (\sin\phi)^{2n} \left(P_{k-n}^{(n,n)}(\cos\phi)\right)^2 \sin\phi\, d\phi = \text{——}$$

—— same calculation ——